Advanced Edge Detection for Noisy SEM Images: A Hybrid Approach for Enhanced Precision

1st Sepuh Melikyan
LEBAX TECHNOLOGIES
Yerevan, Armenia
https://orcid.org/0009-0004-9247-5464

2nd Gurgen Melikian
LEBAX TECHNOLOGIES
Yerevan, Armenia
https://orcid.org/0009-0006-7822-0902

***Abstract*—**
The precision of lithographic processes in microelectronics production is critical, particularly as transistor dimensions continue to shrink. Ensuring the accuracy of these processes relies on robust quality control mechanisms, including effective edge detection in scanning electron microscopy (SEM) images. Traditional image processing techniques often struggle in noisy or low-resolution conditions, and conventional deep learning models can be computationally intensive. This paper introduces a novel hybrid model that combines Holistically-Nested Edge Detection (HED) [9], Lightweight Dense CNN (LDC) [10] and Pixel Difference Network (PiDiNet) [8] to enhance edge detection in challenging SEM imaging scenarios. The proposed model integrates noise suppression and pixel gradient sensitivity methods to achieve precise edge delineation while maintaining computational efficiency. This approach is particularly beneficial for industrial applications such as microcontroller manufacturing, where rapid and accurate SEM image analysis is essential. The source code is released at https://github.com/sepuhhopar/semimage ***Index Terms*—SEM, LDC, PiDiNet, IoU, BCE**

I. Introduction

The development of microelectronics requires precise quality control at every production stage, particularly in the lithographic processes that determine the geometry of semiconductor elements. As transistor dimensions shrink, ensuring the accuracy of the lithography process becomes increasingly essential. Techniques to enhance and validate lithographic quality, especially through analyzing micro-controller structures on scanning electron microscopy (SEM) images [6], have gained prominence. Edge detection on SEM images enables the precise extraction of geometric parameters, essential for lithography model calibration and quantitative quality assessment.

Current SEM image processing methods face limitations in accurately detecting edges under noisy and low-resolution conditions. Traditional approaches lack robustness against such noise, while CNN-based models often require high-resolution images and substantial computational resources. This research proposes a hybrid model combining lightweight architecture and pixel difference convolution for improved edge detection on SEM images. By filling this gap, the study seeks to advance micro-controller manufacturing accuracy through enhanced SEM image analysis capabilities.

The primary goal of the proposed approach is to improve the accuracy of edge detection while maintaining computational efficiency. This approach is especially relevant for industrial applications, such as microcontroller manufacturing, where SEM images need to be processed rapidly and with high precision to ensure quality control.

II. Related Methods

This research utilizes SEM images of microcontrollers of varying sizes and noise levels as the primary material for edge detection. Current methodologies leverage several advanced image processing techniques, integrating both traditional and machine learning-based models for edge extraction.

Traditional edge detection algorithms, such as Sobel [6], Canny [1], and Laplacian filters [4], are tested on SEM images to establish baseline comparisons. However, these classical approaches frequently struggle under noise interference and limited resolution, producing less reliable edge definitions.

Contemporary deep learning models, including Lightweight Dense CNN (LDC) [10] and PiDiNet, play crucial roles in advancing edge detection. LDC is recognized for its lightweight nature, ensuring efficient processing on low-resolution images. Meanwhile, PiDiNet incorporates Pixel Difference Convolution (PDC) [8], which enhances the precision of edge detection by focusing on pixel gradients. The integration of these models into a hybrid architecture enables leveraging LDC's computational efficiency and PiDiNet's gradient sensitivity.

Further enhancing deep learning's role in image recognition, the insights from Simonyan and Zisserman's work on very deep convolutional networks [2] demonstrate that depth, achieved by small convolutional filters, significantly boosts model performance in image classification tasks. This principle underlies modern architectures' use, suggesting that a deep yet computationally efficient model structure can be effective for complex image tasks, such as edge detection in noisy environments.

This hybrid approach addressed the limitations observed in previous methods, providing empirical support for the effectiveness of deep learning-enhanced edge detection in noisy SEM imaging contexts.

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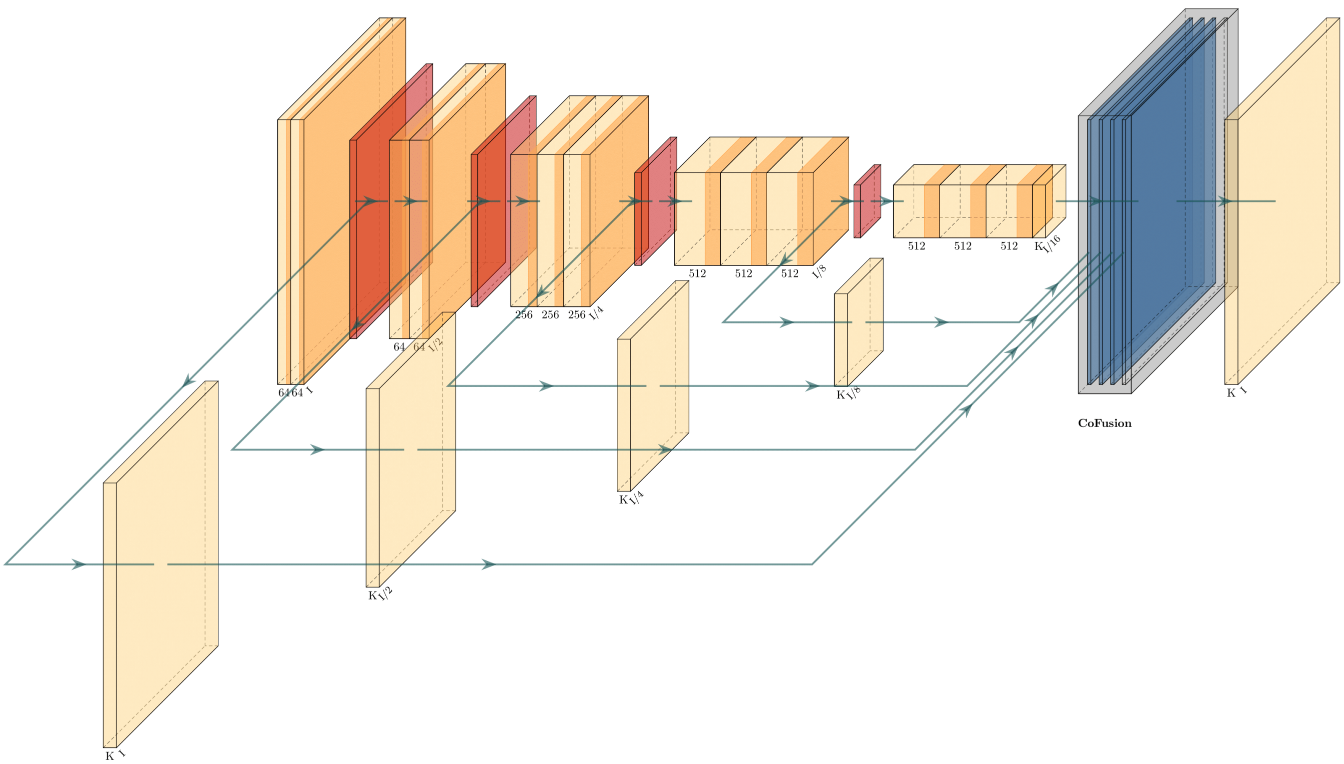


Fig. 1: Architecture of the proposed model.

III. Proposed Approach

The proposed approach focused on developing an optimized model for edge detection in noisy, low-resolution SEM images by combining techniques from three different architectures: Lightweight Dense CNN (LDC) [10] for Edge Detection, Holistically-Nested Edge Detection (HED) [9], and Pixel Difference Networks (PiDiNet), this approach integrates these models into a hybrid architecture, creating a robust system capable of precise edge detection even under challenging image conditions.

The backbone architecture utilized the Holistically-Nested Edge Detection (HED) model [9], known for its strong performance in producing multi-scale edge maps. To enhance this structure, the approach integrated specific techniques from LDC. Specifically, the upscaling and conversion of feature maps into edge maps matching the input image size were accomplished using the USNet module from LDC. Additionally, CoFusion blocks were implemented to merge intermediate edge map predictions. These predictions were fused to create a final, refined output, inspired by the Context-Aware Fusion strategy in CATS. In diagram 1 is illustrated the architecture of proposed method.

Having in mind the noise images as input and detailed output focus of current work, the following modifications to (HED) [9] are proposed to generate the new architecture:

- **Replacement of Standard Convolution**: Standard convolution is replaced with Pixel Difference Convolution (PDC) [8] to enhance edge detection in noisy environments.
- **Noise Suppression Filters**: Filters specifically designed for noise suppression are added to reduce the impact of background noise and improve the clarity of detected edges.

This modification leveraged pixel gradient sensitivity, enabling more precise edge delineation, particularly in noisy environments. These combined strategies resulted in an approach that improved the balance between computational efficiency and detection performance, achieving accurate segmentation in challenging SEM imaging scenarios.

A. *Custom Loss Function*

To optimize training, a custom loss function was designed that balances edge presence detection, boundary accuracy, and noise reduction. This composite loss ensures high-quality edge maps while minimizing errors.

Components of the Loss Function

1) **Binary Cross-Entropy (BCE) Loss for Edge Presence Detection**
   - **Purpose**: Differentiates between edge and non-edge pixels.

- **Weighted for Class Imbalance**: Edge pixels are sparse, so BCE is weighted to focus on subtle edges.
- **Outcome**: Enhances precision and recall for detailed edge capture.

2) **Boundary Loss for Edge Localization**
   - **Purpose**: Enhances edge placement accuracy by minimizing spatial errors.
   - **Method**: Applies a filter to the predicted edge map to highlight edge transitions and penalize deviations.
   - **Outcome**: Ensures detected edges align with actual structures, crucial for SEM image measurements.

3) **Texture Suppression Loss for Noise Reduction**
   - **Purpose**: Reduces background noise and unwanted textures that create false edges.
   - **Method**: Applies a high-pass filter to detect high-frequency noise patterns and penalizes them in non-edge regions.
   - **Outcome**: Produces cleaner edge maps by focusing only on true structural edges.

COMBINED LOSS FUNCTION FORMULATION

The combined loss function L is expressed as:

$$L = \text{BCE} + \alpha \text{ (Boundary Loss)} + \beta \text{ (Texture Suppression Loss)}$$

- **BCE**: Promotes precision and recall in edge detection.
- **Boundary Loss**: Enhances edge localization.
- **Texture Suppression Loss**: Minimizes noise and false edges.

The hyperparameters α and β control the emphasis on boundary accuracy and noise suppression, allowing flexibility for different SEM image characteristics.

IV. TRAINING PROCESS

The training of the hybrid model incorporates a two-stage fine-tuning approach to maximize edge detection performance in SEM images:

A. Two-Stage Training Strategy

1) **Pre-training on General Edge Detection Data**:
   - The model is initially pre-trained on the BSDS500 dataset, a well-known collection of natural images used for edge detection tasks.
   - This phase helps the model learn fundamental edge-detection patterns that provide a solid base for later fine-tuning.

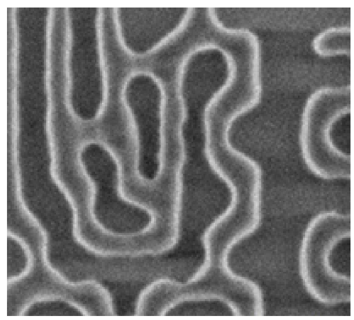
2) **Fine-Tuning on SEM Images**:
   - Following pre-training, the model is fine-tuned using a dataset of SEM images to adapt to the unique noise characteristics and specific edge profiles present in SEM data.
   - A low learning rate is employed during fine-tuning to ensure the retention of general edge-detection features learned in the pre-training phase while adjusting to SEM-specific features.

B. Best Practices for Training

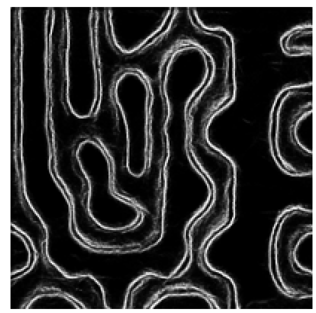
- **Learning Rate Scheduling**: Utilize a learning rate warm-up strategy to prevent initial overshooting and to ensure stable convergence.
- **Data Augmentation**: Apply augmentation techniques such as rotation, noise injection, and contrast adjustment to increase model robustness to real-world SEM image variations.
- **Early Stopping and Regularization**: Implement early stopping to avoid overfitting and add dropout layers in CoFusion to enhance generalization.

V. RESULTS

The proposed hybrid model combining Lightweight Dense CNN (LDC) [10] and Pixel Difference Network (PiDiNet) [8] demonstrated superior performance in edge detection on SEM images when compared to individual implementations of LDC and PiDiNet.



(a) Original SEM Image



(b) Prediction

(c) Ground Truth

TABLE I: Performance Metrics of the Proposed Hybrid Model

Metric	LDC	PiDiNet	Proposed Hybrid Model
Precision	0.85	0.87	**0.91**
Recall	0.82	0.85	**0.89**
F1-score	0.83	0.86	**0.90**
IoU	0.78	0.80	**0.84**
ODS	0.81	0.83	**0.88**
OIS	0.82	0.84	**0.89**

The data in Table I indicate that the proposed hybrid model outperforms both individual LDC and PiDiNet architectures in all evaluated metrics. Notably, the Precision and F1-score improvements highlight the model's ability to maintain high accuracy while effectively reducing noise interference.

A. Performance Highlights

- The hybrid model achieved a Precision of **0.91**, demonstrating its capability to accurately detect edges without significant false positives.
- The Recall score of **0.89** indicates the model's robustness in identifying true edge regions, even under noisy conditions.
- The F1-score and IoU improvements validate the model's balanced performance in edge detection, combining both precision and recall effectively.

The results of this study demonstrate the efficacy of the proposed hybrid model for edge detection in noisy SEM images, leveraging both Lightweight Dense CNN (LDC) and Pixel Difference Network (PiDiNet) [8] architectures. The model's performance was evaluated through a series of quantitative metrics, including precision, recall, F1-score, and Intersection over Union (IoU), along with specific edge detection metrics such as Optimal Dataset Scale (ODS) and Optimal Image Scale (OIS). The evaluation used a set of benchmark images, representative of the typical noise and edge complexity found in SEM imagery of microcontroller components.

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